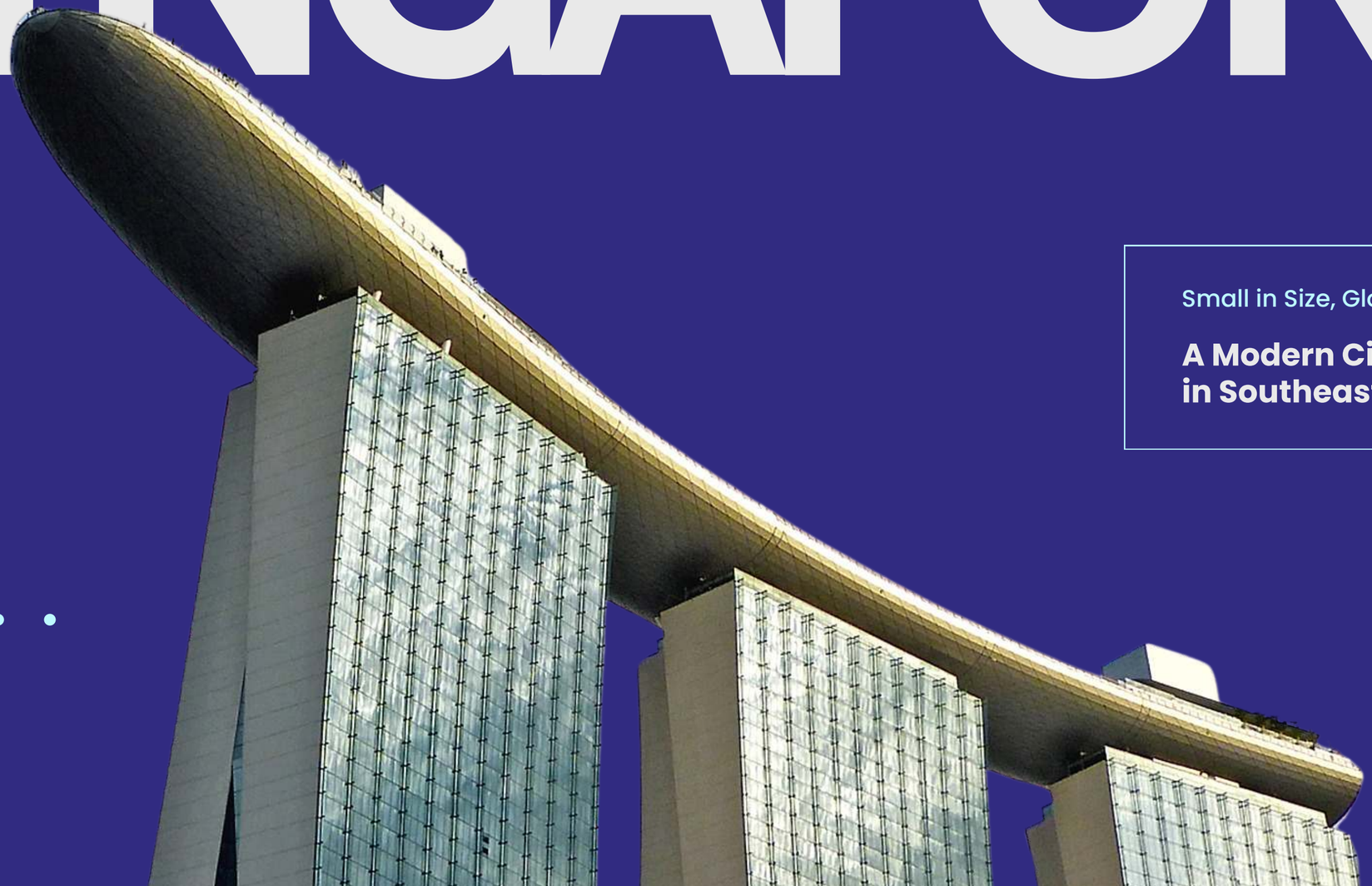
SINGAPORE



Small in Size, Global in Influence

A Modern City-State in Southeast Asia

Presented by

Masterminds

FYBBA || Div- C



OVERVIEW OF *SINGAPORE*



Small Nation, Big Impact

Singapore is a sovereign city-state located at the southern tip of the Malay Peninsula.

It has transformed from a trading port into one of the world's most advanced economies with a high standard of living and global influence.

GEOGRAPHIC *FEATURES*



Compact land area

Despite its limited size, Singapore efficiently uses space through urban planning and vertical development.

Tropical climate

The country experiences warm temperatures and high humidity year-round, with frequent rainfall.

Green city planning

Urban greenery is integrated into infrastructure, creating a clean and sustainable environment.

Environmental Degradation in Earlier Years

- Rapid industrialization in the 1960s–1980s caused heavy pollution
- Factories released smoke and harmful gases into the air
- Rivers became polluted due to industrial and domestic waste
- Fast urbanization reduced green spaces and increased overcrowding
- Poor waste disposal systems affected public health and cleanliness
- Air quality worsened because of traffic growth and industrial emissions

Key Impacts

- Poor AQI and unhealthy living conditions
- Water contamination in rivers and canals
- Increase in respiratory and environmental health issues

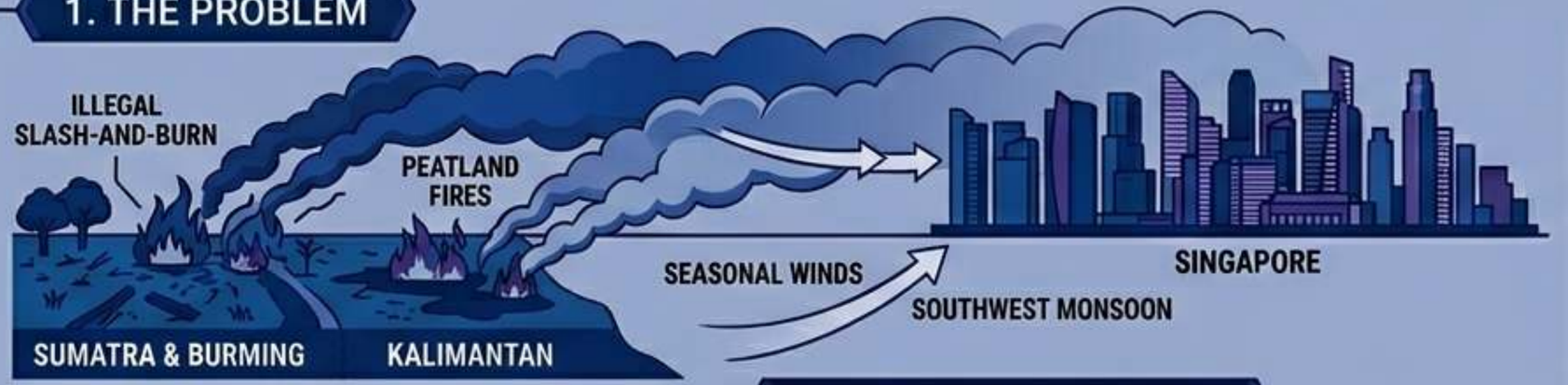
How Singapore Started Recovering

Government Actions & Environmental Reforms

- Strict environmental laws were introduced
- “Clean and Green Singapore” campaign improved public awareness
- Factories were regulated to control emissions
- River-cleaning projects transformed polluted waterways
- Urban planning added parks and green spaces
- Public transport development reduced traffic pollution

SINGAPORE'S TRANSBOUNDARY HAZE PROBLEM: UNDERSTANDING THE CRISIS

1. THE PROBLEM



2. MEASURING AIR QUALITY (PSI)



- GOOD (0-50)
- MODERATE (51-100)
- UNHEALTHY (101-200)
- VERY UNHEALTHY (201-300)
- HAZARDOUS (300+)

PM2.5: KEY POLLUTANT

- GOOD (0-50)
- MODERATE (51-100)
- UNHEALTHY (101-200)
- VERY UNHEALTHY (201-300)
- HAZARDOUS (300+)



3. IMPACTS ON SINGAPORE



PUBLIC HEALTH

- Eye & Throat Irritation
- Asthma Flare-ups
- Cardiovascular Risk



ECONOMY

- Tourism Drops
- Productivity Loss
- Outdoor Events Cancelled



ENVIRONMENT

- Reduced Photosynthesis
- Disturbed Wildlife
- Water Body Contamination

4. MITIGATION & SOLUTIONS



SINGAPORE'S TRANSBOUNDARY HAZE POLLUTION ACT (THPA)



ASEAN COOPERATION



NATIONAL PREPAREDNESS



Source: EVS Project on Singapore Haze

Source: EVS Project on Singapore Haze

Water Scarcity in Singapore

CHALLENGES & HISTORICAL CONTEXT

WHY IS SINGAPORE WATER SCARCE?

🏙️ **No Natural Lakes or Rivers**

City-state with no large freshwater bodies

🌧️ **Rainfall ≠ Self-Sufficiency**

High rainfall but insufficient land to collect it

📈 **Rapid Urbanization**

Population grew from 1M (1960) to 6M+ (2023)

🤝 **Malaysia Dependency**

Historical reliance on Johor for water supply

724
sq km

One of the world's most
water-stressed nations





Singapore (724 km²) | Water demand: 430 million gallons/day

Water Management Solutions

SINGAPORE'S FOUR NATIONAL TAPS STRATEGY

NEWater

Treated reclaimed wastewater using advanced membrane tech. Meets 40% of Singapore's water needs. Ultra-clean, used in industries & top-up reservoirs.

Desalination

Seawater converted to drinking water via reverse osmosis. Two plants supply 30% of needs. Crucial for drought resilience & independence.

Rainwater Harvesting

Two-thirds of Singapore's land used as water catchment. 17 reservoirs collect rain. Marina Barrage turned sea bay into freshwater reservoir.

Imported Water

1962 Water Agreement with Malaysia (Johor). Singapore is reducing dependency; treated & returned some water back. Agreement ends 2061.

40% NEWater share

30% Desalination share

17 Reservoirs

2061 Water treaty end

WASTE MANAGEMENT *& SOIL POLLUTION*

Towards the Zero Waste Singapore

Smart solution for sustainable future



Efficient Waste Management

Singapore Produces large amount of waste due to rapid urbanization & modern lifestyle. To manage this, the country follows "Reduce, Reuse, Recycle" approach. Non-Recycle waste is treated in Waste-to- Energy plants, reducing landfill waste and generating electricity.



Volume Reduction: This process reduces the volume of waste by 90%, significantly extending the lifespan of their only landfill.

WASTE MANAGEMENT *& SOIL POLLUTION*

Towards the Zero Waste Singapore

Smart solution for sustainable future



Sustainable future

Semakau Landfill is a offshore landfill designed to safely manage ash waste while protecting marine ecosystems. Singapore now aims to become a "Zero Waste Nation" through recycling, innovative and waste reduction programs.



Co-existence: It serves as a landfill and a habitat for mangroves and rare seagrasses, proving that waste management can be integrated into nature.

SUSTAINABLE DEVELOPMENT & GREEN INITIATIVES IN SINGAPORE

Building a greener, smarter Singapore

Green Building Initiatives

- Green Mark Scheme
- Energy-efficient buildings
- Rooftop gardens & water-saving systems

Smart City Planning

- Smart traffic systems
- Technology-based city management

Sustainable Development

- Balance between development and environment
- Efficient use of resources



SUSTAINABLE TRANSPORT & ECO-FRIENDLY PROJECTS

Towards a cleaner and greener future

Sustainable Transport

- Extensive MRT network
- Encourages public transport use
- Reduced private vehicle use

Eco-Friendly Projects

- Solar energy projects
- Green spaces and tree plantation
- Vertical gardens

Impact

- Reduced pollution
- Better air quality
- Improved quality of life







CURRENT CHALLENGES & *Climate*

What Singapore faces right now

Rising heat. Rising seas.
Rising emissions.
Singapore faces a triple threat.

+0.24°C	1.15m	53.25 Mt
per decade since 1984 double the global average	sea-level rise by 2100 high emission scenario	CO_2/year — 2nd highest per capita in ASEAN

Rising Temperatures & Extreme Weather

2024 tied as Singapore's hottest year (28.4°C mean). By 2100, temps could rise 0.6°C–5°C more. Urban heat islands push city temps even higher. Flash floods now regularly hit low-lying areas like Orchard Road.

- Very hot days (35°C+) may surge from 4/year → 351/year by 2100 (CCRS V3, 2024)
- Heavy rainfall events increasing — drainage systems under severe strain

Sea-Level Rise & Carbon Emissions

30% of Singapore sits below 5m elevation. Coastal flooding threatens the financial district, homes and fuel facilities. Saltwater intrusion risks contaminating freshwater reservoirs. At 8.9t CO_2/person, Singapore is one of Asia's highest emitters.

- S$100 billion committed over 100 years for coastal protection & flood barriers
- Carbon tax raised to S$25/tonne (2024); targeting S$50–80/tonne by 2030

FUTURE RISKS & *Ecology*

Long-term threats to Singapore's survival

Development makes Singapore prosper — but the environment pays the price.
The future depends on acting now.

37%	>90%	~95%
native species extinct since urbanisation (NUS 2023)	food imported — exposed to global climate shocks	electricity from gas — renewables near impossible

Ecological Imbalance & Biodiversity Loss

Rapid urbanisation has destroyed forests, mangroves and coral reefs. Invasive species are destabilising remaining ecosystems. Singapore has lost most of its original green cover. Coral bleaching is worsening around Sentosa and the Southern Islands.

- Recovery plans for 80+ plant & 40+ animal species — only 40 ha restored in 4 years
- Invasive species like the changeable lizard are crowding out native wildlife

Energy, Food & Water Security

Singapore's climate policy is rated "Highly Insufficient" by Climate Action Tracker 2024 — on track for a 3–4°C world. Limited land makes solar/wind scaling near impossible. Water demand set to double by 2065. Any global climate shock hits Singapore's food supply directly.

- "30 by 30" plan: grow 30% of food locally by 2030 to cut import dependency
- Importing 4GW low-carbon electricity by 2035 + green hydrogen pilot underway

CONCLUSION

Singapore's Environmental Journey

From a polluted industrial city to one of the world's greenest nations — Singapore proves that with strong policy, innovation, and public commitment, environmental recovery is possible.



KEY TAKEAWAYS

Green Policy Works

Strict laws + public campaigns transformed Singapore from smog-filled to globally admired.

Water Innovation

NEWater & desalination turned a water-scarce island into a water self-sufficient nation.

Zero-Waste Vision

Semakau Landfill & waste-to-energy plants show that waste can be minimised and reused.

Smart City Model

Green buildings, efficient transport & urban planning make Singapore a global blueprint.

LESSONS FOR OTHER COUNTRIES

- Invest in clean technology early — prevention is cheaper than cure.
- Combine government policy with community awareness for lasting change.
- Treat water & waste as resources, not problems.

SINGAPORE: BEFORE vs. AFTER

Comparison & Presentation Design

❌ BEFORE — Environmental Crisis

- Heavy air & water pollution from factories
- Singapore River was a dead, foul-smelling drain
- No waste management — open dumping common
- Rapid urbanisation with zero green planning
- Water 100% dependent on Malaysia imports
- Poor AQI — respiratory diseases widespread

VS

✅ AFTER — Green Transformation





- Clean air — strict emission laws enforced
- Singapore River restored; thriving ecosystem
- Semakau Landfill + waste-to-energy plants
- City in a Garden — 47% green cover
- NEWater + desalination = water self-sufficient

CONCLUSION



Singapore improved its environment through smart planning and strict laws.

Sustainable projects helped control pollution and manage resources efficiently.

Singapore is a strong example of balancing development with environmental protection.

Masterminds

THANK YOU!



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